

1. Administrative & Study Identification

Full Study Title: A Phase 2b, Randomized, Active-Controlled, Open-Label Clinical Study to Evaluate a Switch to Islatravir (ISL) and Ulonivirine (ULO) Once Weekly in Adults With HIV-1 Virologically Suppressed on Bictegravir/Emtricitabine/Tenofovir Alafenamide (BIC/FTC/TAF) Once Daily **Short Title/Acronym:** A Study of Islatravir (ISL) and Ulonivirine (ULO) Once Weekly (QW) in Virologically Suppressed Adults With Human Immunodeficiency Virus Type 1 (HIV-1) (MK-8591B-060) **Protocol Number:** MK-8591B-060 **NCT Number:** NCT06891066 **Posting ID:** 425946451014694082 **Sponsor Name:** Merck **Company:** Merck **Investigational Product:** Islatravir (ISL, MK-8591) and Ulonivirine (ULO, MK-8507) **Associated Drug(s):** lactulose, combinations (ATC Code: A06AD61) **Phase of Development:** PHASE2 **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Medical Monitor:** Clinical Director, Merck Sharp & Dohme LLC

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of a switch to once-weekly islatravir (ISL) and ulonivirine (ULO) by determining the proportion of participants who have a high level of the virus in their blood (HIV-1 RNA ≥ 50 copies/mL) after 24 weeks, compared to an existing daily treatment (BIC/FTC/TAF). To also evaluate safety by tracking adverse events (AEs) and discontinuations due to AEs. **Secondary Objectives:** To evaluate the safety, tolerability, and pharmacokinetics of the MK-8591B (ISL + ULO) once-weekly regimen in adults with HIV-1.

2.2 Background & Rationale To address the need for better and more convenient treatments for people with HIV-1. Daily oral antiretroviral therapy (ART) can be difficult for some patients to maintain long-term. This study investigates whether a new combination treatment (ISL + ULO) taken once a week can maintain viral suppression as effectively as the daily standard of care (BIC/FTC/TAF), potentially reducing the pill burden and improving long-term adherence.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (BIC/FTC/TAF) **Blinding:** Open-label **Randomization:** Randomized **Phase Requirement:** Trial must be in PHASE2 (Phase ID: PHASE2)

3.2 Planned Enrollment & Duration Targeted Enrollment: 150 participants **Number of Sites:** Global (Locations include United States, Australia, Puerto Rico, and Switzerland) **Estimated Study Start:** 14-Apr-2025 **Estimated Primary Completion:** 10-Aug-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years old. 2. Has been receiving Bictegravir/Emtricitabine/Tenofovir alafenamide (BIC/FTC/TAF) therapy with documented viral suppression [Human immunodeficiency virus type 1 (HIV-1) ribonucleic acid (RNA) < 50 copies/mL] for ≥ 6 months prior to providing documented informed consent and has no history of prior virologic treatment failure on any past or current regimen.

4.2 Exclusion Criteria 1. Has Human immunodeficiency virus type 2 (HIV-2) infection. 2. Has a diagnosis of an active Acquired immune deficiency syndrome (AIDS)-defining opportunistic infection. 3. Has active hepatitis C virus (HCV) coinfection or hepatitis B virus (HBV) coinfection. 4. Has a history of malignancy ≤ 5 years prior to providing documented informed consent except for adequately treated basal cell or squamous cell skin cancer, in situ cervical or in situ anal cancer, or cutaneous Kaposi's sarcoma. 5. Has prior exposure to Islatravir (ISL) or Ulonivirine (ULO) for any duration any time prior to Day 1.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Investigational Arm: ISL 2 mg + ULO 200 mg administered once a week (QW). Comparator Arm: BIC/FTC/TAF administered once every day. **Route of Administration:** Oral (Capsules/Tablets) **Duration of Treatment:** Part 1 involves 48 weeks of treatment; Part 2 continues treatment from week 49 through week 96.

5.2 Concomitant & Prohibited Therapy Permitted: Standard allowed medications not interacting with NNRTI or NRTTI pathways. (May include lactulose, combinations; ATC: A06AD61 where applicable). **Prohibited:** Immunosuppressive therapies, medications that significantly induce or inhibit CYP enzymes affecting the metabolism of ULO or ISL, or concurrent investigational agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Viral Load (HIV-1 RNA), CD4 count, Labs
Baseline	Day 0	Randomization, First Dose of Study Drug	
Interim	Week 24 $\pm$ 7	Efficacy Assessment (Viral load	

check), Safety Labs, AE Monitoring | | **End of Part 1** | Week 48 \$
\pm\$ 7 | Viral Load Check, Safety Labs, Treatment Continuation
Decision | | **End of Study**| Week 96 | Final Viral Load Assessment,
Exit Interview, IP Accountability |

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

- 1. Percentage of Participants With HIV-1 RNA ≥ 50 copies/mL at Week 24:** Plasma HIV-1 ribonucleic acid (RNA) quantification will be performed at the central laboratory using a polymerase chain reaction (PCR) assay. Percentage of participants with HIV-1 RNA ≥ 50 copies/mL will be reported at week 24.
- 2. Percentage of Participants who Experience an Adverse Event (AE):** An AE is any untoward medical occurrence in a participant, temporally associated with the use of study treatment, whether or not considered related to the study treatment. The percentage of participants who experience an AE will be reported.
- 3. Percentage of Participants Discontinuing Study Treatment due to AEs:** An AE is any untoward medical occurrence in a participant, temporally associated with the use of study treatment, whether or not considered related to the study treatment. The percentage of participants who discontinue study treatment due to an AE will be reported.

7.2 Secondary Outcomes

- 1. Percentage of Participants With HIV-1 RNA ≥ 50 copies/mL at Week 48**
- 2. Percentage of Participants With HIV-1 RNA < 50 copies/mL at Week 24**
- 3. Percentage of Participants With HIV-1 RNA < 50 copies/mL at Week 48**

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via patient reporting and clinical laboratory assessments (e.g., monitoring total lymphocyte and CD4 counts). Percentage of participants who experience an AE and those discontinuing due to AEs are evaluated as primary outcomes. **Serious Adverse Events (SAEs):** Standard reporting timeline, typically within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** External, independent Data Monitoring Committee (eDMC) established by Merck to monitor patient safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for all participants who receive at least one dose of study medication. **Sample Size Justification:** Powered to demonstrate non-inferiority of the ISL+ULO once-weekly regimen compared to daily BIC/FTC/TAF. **Handling of Missing Data:** FDA Snapshot algorithm (missing = failure) will be primarily used for virologic outcome measures at Week 24 and Week 48.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring visits to ensure protocol adherence and data integrity. **Audit Trail:** Electronic Case Report Form (eCRF) entry with full audit trails, strict data clarification forms, and central lab data reconciliation.

11. Study Identifiers & Governance

Primary ID: MK-8591B-060 **Registry Number:** NCT06891066 **Sponsor/Lead Organization:** Merck **Study Title:** A Study of Islatravir (ISL) and Ulonivirine (ULO) Once Weekly (QW) in Virologically Suppressed Adults With Human Immunodeficiency Virus Type 1 (HIV-1) (MK-8591B-060) **Official Regulatory Title:** A Phase 2b, Randomized, Active-Controlled, Open-Label Clinical Study to Evaluate a Switch to Islatravir (ISL) and Ulonivirine (ULO) Once Weekly in Adults With HIV-1 Virologically Suppressed on Bictegravir/Emtricitabine/Tenofovir Alafenamide (BIC/FTC/TAF) Once Daily

12. Clinical Framework (Disease Information)

Disease Areas: Human Immunodeficiency Virus Type 1 (HIV-1) Infection **Preferred UMLS Name:** Human Immunodeficiency Virus Type 1 Infection **Name:** HIV **Preferred Name:** HIV Infections **Semantic Type:** Disease or Syndrome **Definition:** An infection caused by the human immunodeficiency virus. **MeSH Code:** D015658 **SNOMED ID:** 19030005 **Diagnosis Threshold:** Virologically suppressed (HIV-1 RNA < 50 copies/mL) for ≥ 6 months **Medical Methodology:** Antiretroviral Therapy (ART) Switch **Optimization Target:** Maintenance of viral suppression on a once-weekly dosing schedule

13. Study Procedures & Methodology

Management Pathway: Switch from standard daily oral ART to once-weekly ISL/ULO regimen **Education Protocol (HCP):** Training on protocol-specific dispensing, CD4/lymphocyte monitoring, and handling virologic rebound. **Education Protocol (Patient):** Adherence counseling for the once-weekly dosing routine. **Quality Audit Mechanism:** Viral load tracking and adherence monitoring (pill counts).

14. Observation & Follow-Up Schedule

Total Duration: 96 weeks of active treatment phase **Onsite Visit Cadence:** Routine visits spanning Baseline, Weeks 4, 12, 24, 48, 72, and 96. **Remote Monitoring Frequency:** Between onsite visits or as indicated by standard ART monitoring. **Checklist Review Interval:** Routine check-ins for AE monitoring and adherence.

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					